



An automated classification method for acute leukemia in blood smears based on microscopic hyperspectral imaging and multimodal deep learning

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ABSTRACT

Manual microscopic examination of blood smears presents reliability and efficiency challenges. Existing automated methods predominantly rely on whole-image analysis, limiting the quantitative assessment of blast cell proportions. This study proposes an automated framework integrating microscopic hyperspectral imaging (MHSI) with deep learning to precisely classify blood smears as normal, acute myeloid leukemia (AML), or acute lymphoblastic leukemia (ALL). We acquired hyperspectral data from 110 blood smears, applied U²-Net to segment individual cells, and for each cell extracted both spectral features and 63 phenotypic features (morphology, texture, color). To address class imbalance, we introduced data augmentation using a Wasserstein generative adversarial network with gradient penalty (WGAN-GP), and found that two one-dimensional convolutional neural network (1D-CNN) models (1D-GoogLeNet and 1D-ResNet) both achieved their best performance on a Balanced dataset in which class distributions were equalized. For spectral features, 1D-ResNet performed best, yielding a cellular-level classification accuracy of 89.52 %, while for phenotypic features, 1D-GoogLeNet performed best, with an accuracy of 89.05 %. Based on the 1D-ResNet and 1D-GoogLeNet architectures, we further conducted a systematic evaluation of three multimodal fusion strategies: data-level fusion, feature-level fusion, and decision-level fusion. The results showed that the data-level fusion model achieved a cellular-level classification accuracy of 85.24 %, and the feature-level fusion model achieved 89.52 %, whereas decision-level fusion outperformed both, with the best model reaching a cellular-level classification accuracy of 94.76 %. In an external validation set of 15 independent clinical samples, the model achieved 92.90 % accuracy at the cellular level and, by quantifying the proportions of different white blood cell (WBC) types, reached 100 % accuracy at the smear level. This study demonstrates that integrating MHSI technology with quantitative cell proportion analysis provides an efficient and accurate automated approach for leukemia diagnosis, offering new insights into leukemia classification.

1. Introduction

Acute leukemia (AL) is a malignant clonal disease originating from hematopoietic stem cells, characterized by excessive proliferation of abnormal blast cells in the bone marrow, leading to severe impairment of normal hematopoietic function [1]. Based on pathological and immunophenotypic characteristics, acute leukemia is predominantly classified into acute myeloid leukemia (AML) and acute lymphoblastic

leukemia (ALL) [2]. Given the substantial differences between AML and ALL regarding treatment strategies, clinical manifestations, and prognosis, accurate differentiation among AML, ALL, and normal blood smears is of critical importance for clinical decision-making and prognostic assessment [3].

Currently, clinical diagnosis of acute leukemia mainly relies on morphological examination, immunophenotypic analysis, molecular genetic testing, and cytogenetic evaluation [4]. Among these methods,

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morphological examination of blood smears serves as the fundamental diagnostic approach, typically requiring hematopathologists to classify cells manually under a microscope. However, this method has significant drawbacks. First, diagnostic results heavily depend on the expertise and clinical experience of pathologists, resulting in variability and reduced repeatability due to subjective factors. Second, the method is labor-intensive and time-consuming, failing to meet the demands for efficient and precise clinical diagnosis [5]. While immunophenotyping, molecular genetic analysis, and cytogenetic evaluations offer higher accuracy, they are complex, expensive, and require lengthy turnaround times, limiting their utility for large-scale rapid screening [6–8]. Therefore, developing a computer-aided automated diagnostic system for accurate classification of blood smear types is of significant clinical value.

In recent years, rapid advances in computer vision and artificial intelligence have facilitated the wide application of deep learning methods, such as convolutional neural networks (CNNs), in hematological image analysis. Current studies primarily follow two approaches. First, most studies directly input white blood cell (WBC) microscopic images into deep learning models for classification of different WBC types in blood smears [9–11]. Although these methods exhibit excellent accuracy, their interpretability remains limited due to difficulties in clarifying how morphological, textural, and color-based cellular phenotypic features contribute to classification. Additionally, most studies focus solely on primary WBC types from normal smears and cannot effectively discriminate among normal, AML, and ALL smears, thus restricting their clinical applicability. Second, some groups have attempted automated acute leukemia classification based on microscopic image data. For instance, Abhishek et al. constructed a dataset containing microscopic images of normal, AML, and ALL smears, employing various pretrained models for transfer learning, and achieved an accuracy of 96.67 % on heterogeneous datasets [12]. Jawahar et al. proposed a lightweight deep convolutional neural network called ALNett for ALL classification, achieving 91.13 % accuracy and a 0.96 F1-score on the C-NMC 2019 dataset [13]. Claro et al. introduced the “Alert Net-RWD” model, applying 20-fold data augmentation on 2415 blood smear images, and achieved 97.18 % accuracy in distinguishing normal, AML, and ALL smears [14]. Despite these significant advancements, several prominent issues remain unresolved. First, deep learning models generally require extensive datasets and considerable computational resources, posing challenges for clinical implementation. Second, these methods typically classify entire images directly, neglecting quantitative analysis of specific cell types and proportions, thus deviating from clinical morphological diagnostic criteria, limiting their acceptance by clinicians. Finally, current methods lack in-depth exploration of cellular biochemical properties. Therefore, there is an urgent need for a more precise and interpretable approach that incorporates cell proportion quantification to differentiate normal, AML, and ALL blood smears effectively.

Microscopic hyperspectral imaging (MHSI), integrating spectral analysis and digital imaging, simultaneously acquires cellular images and spectral information [15,16]. Utilizing MHSI for blood smear data collection, coupled with cellular phenotypic features and spectral information extraction, provides a feasible strategy for cell classification. Compared to image-based deep learning alone, cellular phenotypic feature extraction more intuitively captures structural and functional cellular changes, reduces the parameter demands of deep learning models, and simplifies training. However, leukemia datasets often exhibit class imbalance, especially involving limited rare cell populations; thus, data augmentation using generative adversarial networks (GANs) to alleviate imbalance is critically important. GANs and their variants, such as DCGAN and Wasserstein GAN (WGAN), have been extensively employed in data generation [17–19]. Moreover, due to complementary information between spectral and phenotypic data, exploring multimodal fusion strategies at data-level, feature-level, and decision-level can effectively exploit the advantages of different data

types, further enhancing classification accuracy.

Based on the above considerations, this study proposes an automated acute leukemia classification framework integrating microscopic hyperspectral imaging and deep learning to overcome limitations of conventional methods regarding cellular biochemical feature extraction and classification accuracy. The workflow is illustrated in Fig. 1. Specifically, the study involved hyperspectral data collection from blood smears, synchronous extraction of cellular spectral and phenotypic features, and class imbalance mitigation using WGAN-GP. Multilevel fusion strategies were explored to optimize model performance. Ultimately, the optimal model was applied to classify cells and quantify the number and proportion of different cell types, thereby enabling accurate automated differentiation among normal, AML, and ALL blood smears. To our knowledge, this is the first study integrating microscopic hyperspectral imaging with deep learning to classify blood smears based on cell proportion changes. The proposed approach exhibits significant innovation and clinical potential, expected to substantially enhance the efficiency and accuracy of acute leukemia diagnosis and provide robust technological support for clinical diagnosis and treatment.

2. Methods

2.1. Ethics statement

This study was approved by the Institutional Review Board of the First Affiliated Hospital of Jinan University (KY-2025-040). Informed consent was waived due to the retrospective nature of the study and the anonymization of the analyzed data.

2.2. Preparation of blood smear samples

Blood smear samples used in this study were provided by the First Affiliated Hospital of Jinan University. Samples were collected from 110 subjects, comprising 41 healthy individuals, 38 patients diagnosed with AML, and 31 patients diagnosed with ALL. All samples underwent routine Wright staining to ensure clear cellular morphology. Before hyperspectral data acquisition, blood smears were stored in sealed microscope slide boxes under dark, dry, and temperature-controlled conditions to maintain staining quality and sample stability. Additionally, due to uneven surfaces typically observed in blood smears, pathologists placed a coverslip onto each smear surface, adhering it with neutral balsam, thus ensuring samples remained in a uniform plane to facilitate accurate hyperspectral data acquisition.

2.3. Acquisition of hyperspectral data and color images

Hyperspectral data were acquired using a microscopic hyperspectral imaging system based on an inverted microscope (ECLIPSE Ti-U, Nikon). The system was equipped with a full-color CCD camera mounted on the left port, and a digital CMOS camera (ORCA-Flash4.0 LT Digital CMOS Camera, C11440-42U) coupled with a liquid crystal tunable filter (VariSpec VIS, CRI Inc.) mounted on the right port. Illumination was provided by a halogen-tungsten lamp. Imaging was performed across a wavelength range of 420–720 nm, with a step size of 2 nm, covering 151 spectral bands. The exposure time was set at 0.01 ms with a magnification of 150 times. This spectral range was selected based on previous evidence indicating its efficacy in characterizing biochemical differences in cells and its successful application in leukemia cell classification [20–23].

During data acquisition, each blood smear sample was first fixed onto the stage of the microscopic hyperspectral imaging system. The regions of interest were identified by microscopic observation through the eyepiece. Subsequently, microscopic hyperspectral images and corresponding color images were captured from the same field of view. To facilitate spectral calibration, microscopic hyperspectral images of blank microscope slides were also acquired under identical imaging

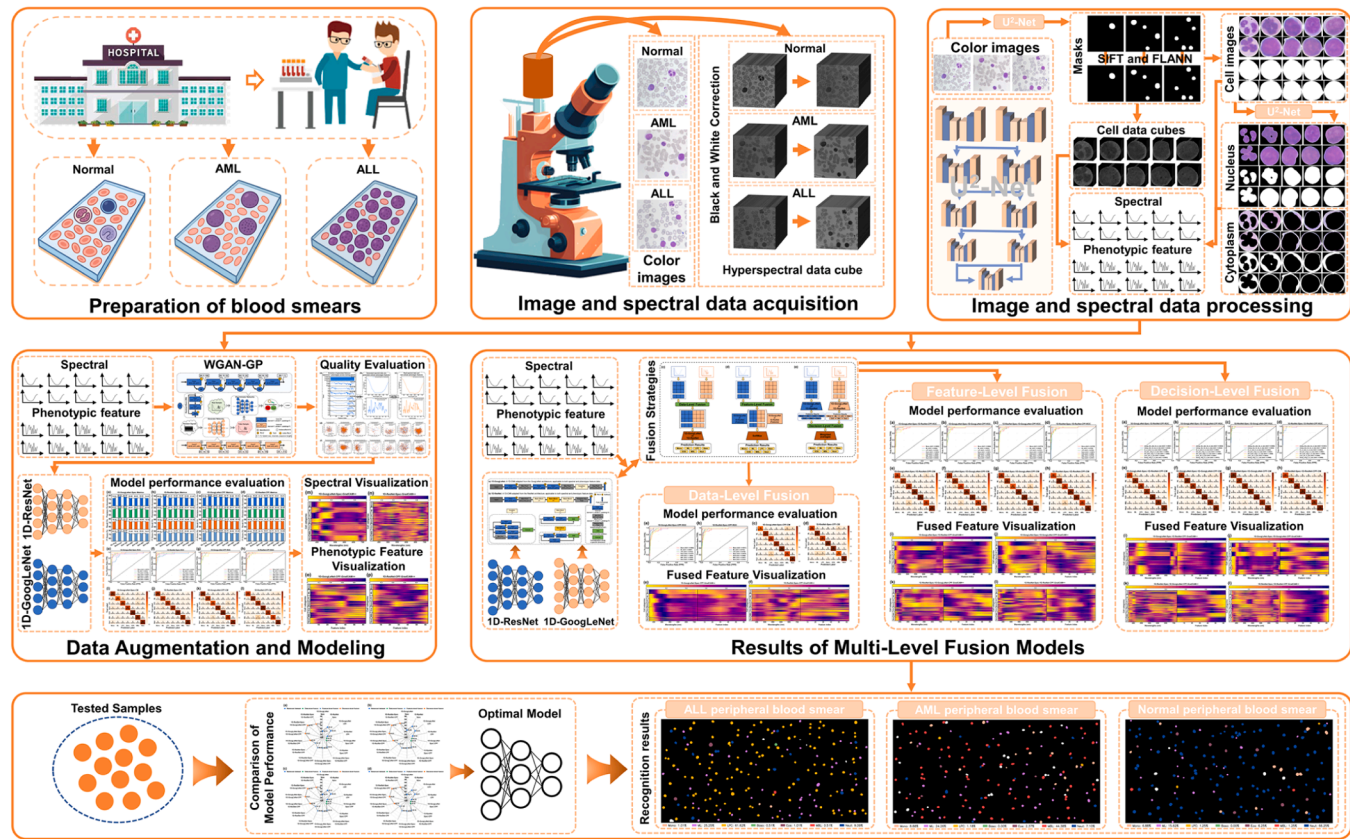


Fig. 1. Graphical workflow of acute leukemia diagnosis based on MHSI and multimodal deep learning.

conditions.

2.4. Data processing and cell annotation

In microscopic hyperspectral imaging technology, single-wavelength images are typically obtained as grayscale images. Such images often exhibit blurred cell boundaries and insufficient contrast between the cytoplasm and background regions, making it challenging to directly

segment cells accurately. In contrast, simultaneously captured color microscopic images exhibit clearer and more distinct cell boundary information. Therefore, in this study, color images were utilized to assist in the segmentation of grayscale single-wavelength images, enabling accurate localization and precise segmentation of target cells (for detailed implementation, see [Supplementary Materials](#)).

All segmented cells were manually annotated by experienced pathologists. According to the annotation results, the hyperspectral data

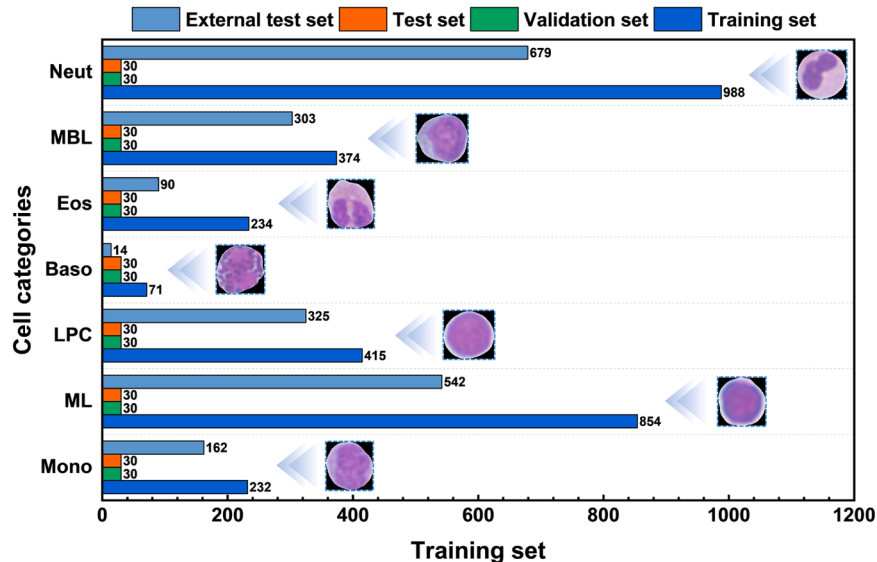


Fig. 2. Cell Count Statistics for Different Cell Types. Note: Neut = Neutrophil; MBL = Myeloid blast; Eos = Eosinophil; Baso = Basophil; LPC = Lymphoid precursor cell; ML = Mature lymphocyte; Mono = Monocyte.

and corresponding color images for each cell were saved into their respective independent folders. Subsequently, the spectral data and phenotypic features of each cell were extracted (see [Supplementary Materials](#) for detailed procedures) for subsequent modeling and analysis. The detailed statistics of the annotated cells are presented in [Fig. 2](#). In total, 5703 cells were annotated in this study. Cells from 15 blood smear samples were separately designated as the external validation set to assess the generalization performance of the model in distinguishing between normal, AML, and ALL cells. The remaining cells were divided into training, validation, and test sets with proportions of 3168, 210, and 210 cells, respectively.

2.5. Basis for distinguishing normal, AML, and ALL blood smears

The early hallmark of acute leukemia is clonal proliferation and maturation arrest occurring at early differentiation stages of hematopoietic precursor cells, resulting in an abnormal release of large numbers of blast cells from the bone marrow into peripheral blood, thereby disrupting the normal balance of hematopoietic lineages [24,25]. Consequently, the steady state, typically maintained by mature granulocytes, mature lymphocytes, erythroid cells, and monocytes, is disrupted and replaced by blast cells of a single lineage. This manifests as a distinct imbalance in cell proportions observed in blood smears, providing direct, quantifiable evidence for hematological diagnosis.

According to the WHO Classification of Haematolymphoid Tumours, normal peripheral blood typically comprises 40–60 % mature neutrophils, 20–40 % mature lymphocytes, and less than 10 % other granulocytes and monocytes combined. A significant increase in myeloblast proportions within blood smears, typically reaching or exceeding 20 %, serves as an essential diagnostic criterion for AML. Similarly, an increase in lymphoid precursor cells above 20 % indicates ALL [26,27].

Thus, by accurately identifying and counting each cell type in blood smears using deep learning models, and subsequently assessing blast cell proportions according to WHO thresholds, automatic and highly accurate differentiation of normal, AML, and ALL blood smears can be reliably achieved.

3. Results

3.1. Spectral analysis of different cell types

[Fig. 3](#) illustrates the transmission spectra of seven WBC types across the wavelength range of 420–720 nm. Each spectral curve represents the average transmittance of the corresponding cell type at each wavelength, reflecting selective absorption and scattering behaviors of intracellular components. The shaded areas denote standard deviations, indicating variability in spectral measurements. Overall, the spectral curves exhibited good reproducibility within each cell type, characterized by relatively small standard deviations, suggesting low intracellular compositional heterogeneity and high spectral consistency.

In general, the transmission spectra for all cell types displayed a characteristic “U-shaped” pattern, indicating low transmittance within the 450–650 nm band and a subsequent increase above 650 nm. This trend likely reflects common biochemical or structural properties across various cell types. Notably, substantial differences in the spectral curves among cell types were observed within the wavelength range of 500–650 nm, demonstrating clear spectral separability. Specifically, basophils exhibited significantly lower transmittance in this wavelength region compared to other cell types, likely due to abundant basophilic granules within their cytoplasm, characterized by intense absorption and scattering of incident visible light. Conversely, eosinophils showed a distinct peak within the 530–600 nm band, attributed to the eosinophilic granules in their cytoplasm that strongly bind eosin dye, resulting in an orange-yellow optical appearance that increases transmittance. In contrast, other cell types demonstrated relatively stable and less distinctive spectral changes within this range.

3.2. Classification models based on spectral data and cellular phenotypic features

To investigate the classification performance of the 1D-GoogLeNet and 1D-ResNet models using cell spectral and phenotypic feature data with varying dataset sizes, seven datasets were prepared: No Aug, Balanced, and Bal+ 100 through Bal+ 500 (the data augmentation process is detailed in the [Supplementary Materials](#)). [Fig. 4\(a–d\)](#) illustrate performance metrics comparing the 1D-GoogLeNet and 1D-ResNet models across spectral and phenotypic datasets with different sizes.

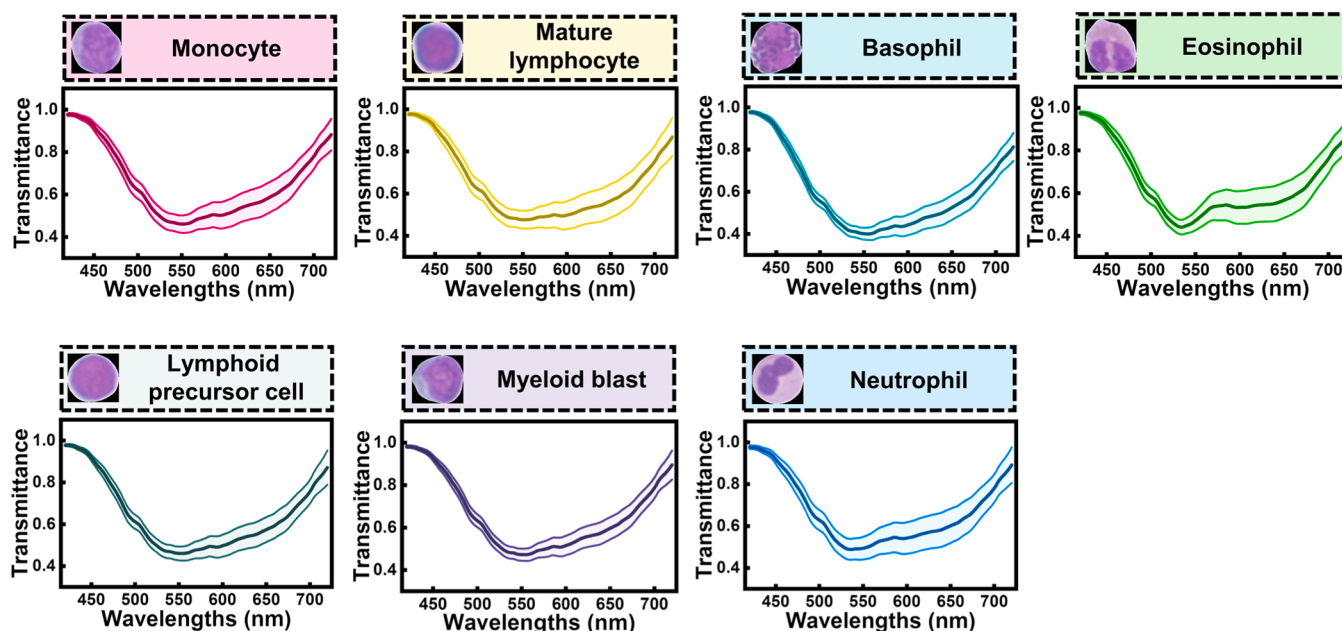


Fig. 3. Average spectral diagrams of different cell types.

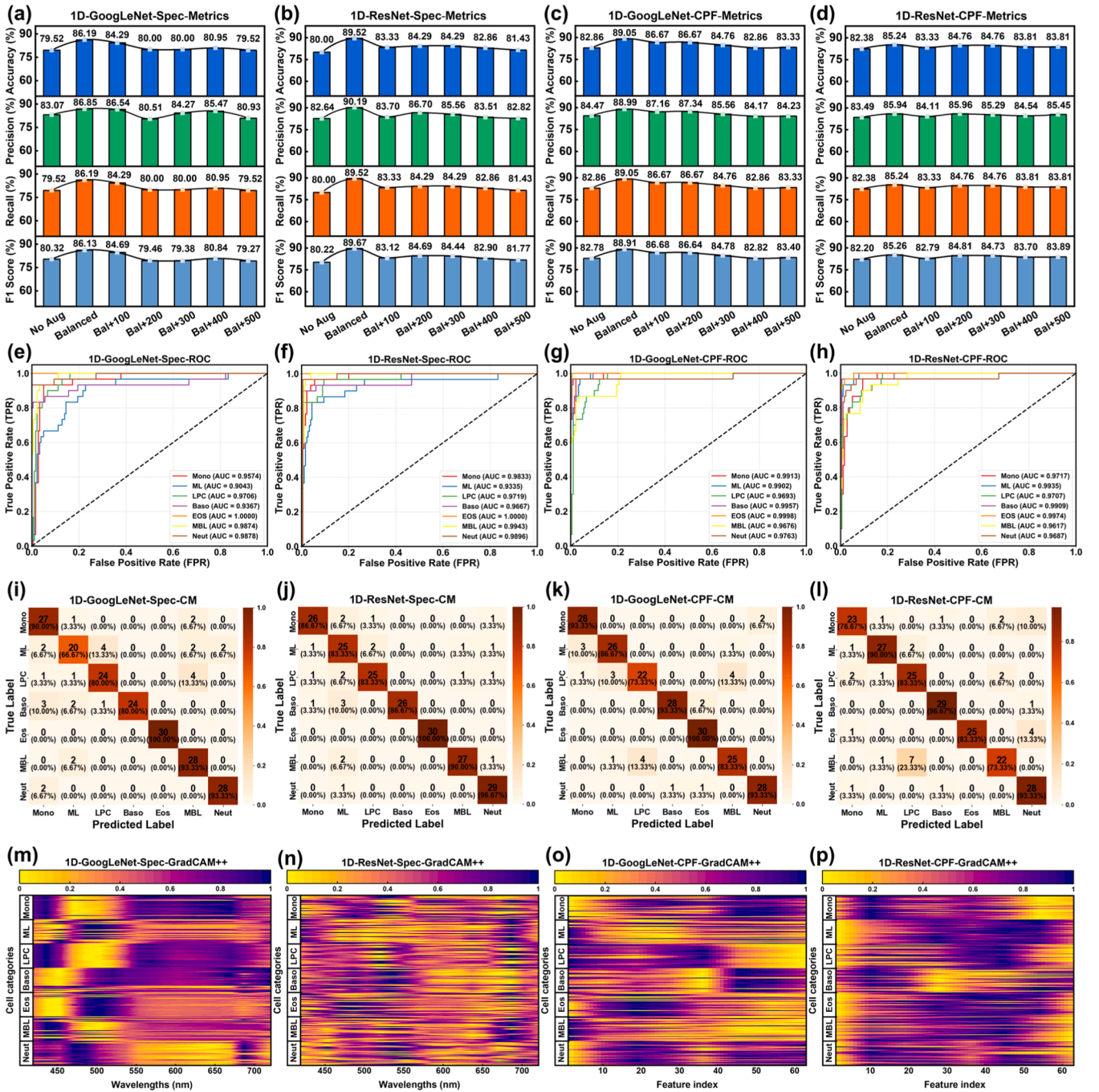


Fig. 4. Classification performance and visualization of models based on spectral and phenotypic data of cells, (a–d) present the performance comparisons of 1D-GoogleNet and 1D-ResNet on spectral and phenotypic datasets of varying scales. (e–h), (i–l), and (m–p) show the multiclass ROC curves, confusion matrices, and GradCAM++ heatmaps of four models trained on the balanced dataset, respectively. Note: “No Aug” refers to the original dataset without augmentation; “Balanced” indicates a class-balanced dataset with each class augmented to match the size of the majority class; “Bal+ 100”, “Bal+ 200”, ..., “Bal+ 500” represent balanced datasets where each class is further augmented by 100, 200, ..., and 500 additional samples, respectively. Abbreviations are defined in the legend of Fig. 2.

Both models achieved their highest accuracy, precision, recall, and F1-score on the Balanced dataset, clearly outperforming datasets with no augmentation or excessive expansion. This observation suggests that moderate data augmentation (such as the Balanced dataset) effectively mitigates class imbalance and enhances model performance. However, excessive augmentation reduced model accuracy, mainly because abundant synthetic samples introduced distribution shifts and irrelevant information, thereby compromising generalization and potentially causing overfitting. This finding aligns with previous studies indicating diminishing returns from data augmentation, with excessive reliance on

synthetic data potentially weakening model performance on real datasets [28,29].

As shown in Fig. 4(e–h), both models exhibited strong performance, with the AUC values for all cell types exceeding 0.9000. Among these, eosinophils demonstrate the most outstanding performance in classification based on spectral data, achieving an AUC value of 1.0000. This result is primarily attributed to the unique transmission peak of eosinophils in the 530–600 nm wavelength range, which provides the model with a strong discriminative signal. Additionally, the AUC values for monocytes, lymphoid precursor cells, and neutrophils all exceed 0.9500,

indicating that the model has strong recognition capabilities for these cell types. In classification based on cellular phenotypic feature data, the classification performance for monocytes, basophils, and eosinophils was particularly notable, with AUC values all exceeding 0.9900. Compared to spectral data, both models demonstrated significantly improved accuracy in identifying mature lymphocytes, indicating that phenotypic features possess stronger discriminative capability in the identification of this cell type.

Fig. 4(i–l) further indicate that neither model exhibited severe bias towards any particular cell class, reinforcing the importance of balanced datasets for enhancing classification performance. Under the spectral data condition, both models demonstrated high classification accuracy. Notably, eosinophils achieved the highest identification accuracy, reaching or approaching 100.00 %, while myeloid blasts and neutrophils also exhibited strong classification performance, with accuracies exceeding 90.00 %. However, in the 1D-GoogLeNet model, the classification accuracy for mature lymphocytes was relatively low at 66.67 %, due to frequent misclassifications with monocytes and lymphoid precursor cells. In contrast, the 1D-ResNet model showed improved performance in this regard, increasing mature lymphocytes classification accuracy to 83.33 %. Under the phenotypic feature condition, basophils and neutrophils were classified with high accuracy, both exceeding 93.00 %. Nevertheless, lymphoid precursor cells remained prone to misclassification. Additionally, the 1D-ResNet model demonstrated particularly strong classification performance for mature lymphocytes, achieving an accuracy of 90.00 %, although a slight decline in accuracy was observed for monocytes and myeloid blasts.

From Fig. 4(m–p), both 1D-GoogLeNet and 1D-ResNet models successfully identified discriminative feature regions within specific wavelength bands or feature intervals, with differences observed between models in terms of highlighted areas. For spectral data, the attention heatmaps from both models closely matched known characteristic spectral features of each cell type. For instance, they both emphasized the low transmittance bands of basophils and the characteristic peak of eosinophils. For phenotypic data, significant consistency between the two models was observed only in neutrophils, while other cell types showed nearly mutually exclusive regions of attention, resulting in differences in model performance. For example, eosinophils exhibited substantial cytoplasmic feature differences compared to other cells, and the 1D-GoogLeNet model identified a highly discriminative

region within features 18–24, corresponding primarily to cytoplasmic texture derived from color co-occurrence matrices. Consequently, 1D-GoogLeNet achieved superior EOS classification accuracy compared to 1D-ResNet. Overall, the 1D-GoogLeNet, utilizing multi-scale parallel convolutional structures via the Inception module, excelled in extracting local and fine-grained features. Conversely, 1D-ResNet exhibited more balanced feature responses, covering broader high-weight areas across multiple cell types, highlighting its advantage in capturing global features.

In summary, systematic data augmentation experiments and multi-model comparisons confirmed that class balancing is critical for enhancing cell classification performance. Different data modalities provided varying classification efficacy for WBCs, suggesting the potential for further exploration of multimodal data fusion strategies to improve accuracy and robustness.

3.3. Classification models with data-level fusion

Fig. 5(a) and (b) reveal that data-level fusion did not universally outperform single-modality models in terms of AUC values; in fact, classification performance for certain cell types was somewhat reduced. This phenomenon may result from redundant information introduced during fusion, ineffective exploitation of complementary features, or suboptimal model adaptability to high-dimensional fused data. Nevertheless, both deep-learning models exhibited generally strong classification performance across various WBC types. The 1D-GoogLeNet model achieved perfect classification performance for eosinophils, with an AUC value of 1.0000, and demonstrated high classification capability for other cell types as well, with all AUC values exceeding 0.9550. Similarly, the 1D-ResNet model demonstrated consistently strong performance, achieving AUC values above 0.9610 for all cell subtypes, indicating robust model behavior.

Fig. 5(c) and (d) demonstrate that the 1D-GoogLeNet model achieved high accuracy for most cell types, notably achieving 100 % and 96.67 % accuracy for eosinophils and neutrophils, respectively. However, the monocyte class exhibited the lowest classification accuracy at only 63.33 %, with most misclassifications occurring in the myeloid blast and neutrophil categories. The classification performance of the 1D-ResNet model was comparable to that of 1D-GoogLeNet, with basophils, eosinophils, and neutrophils all achieving an accuracy of 96.67 %.

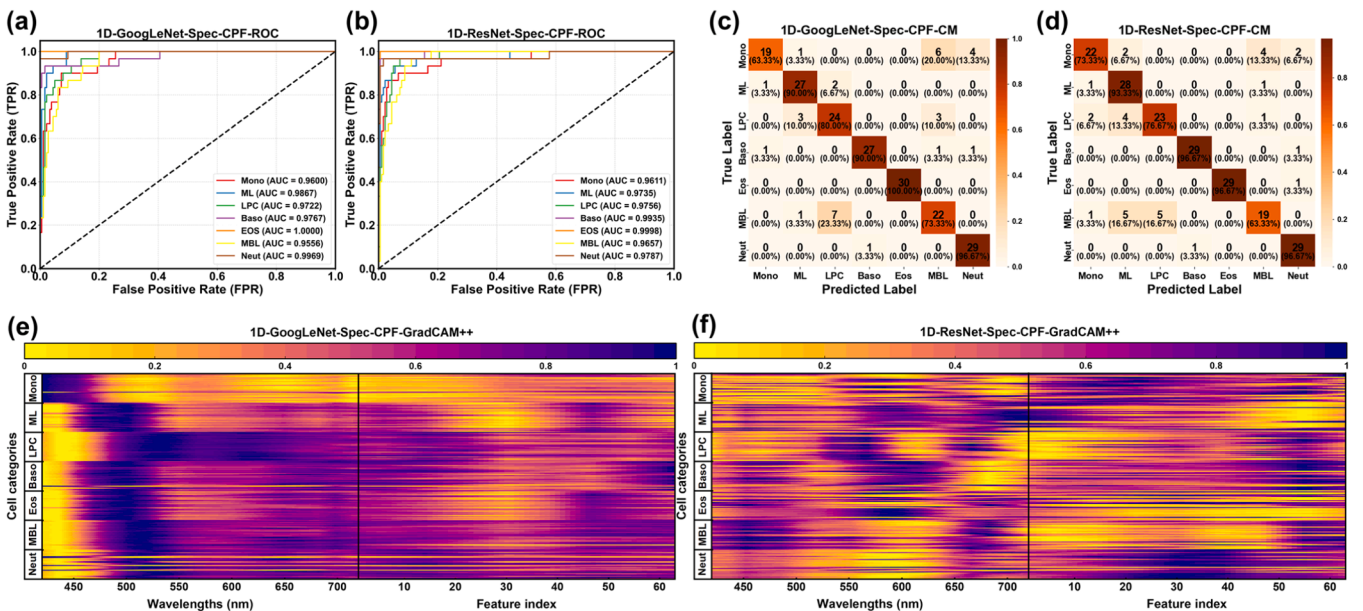


Fig. 5. Classification performance and visualization of models based on data-level fusion strategy, (a–b), (c–d), and (e–f) present the multiclass ROC curves, confusion matrices, and GradCAM++ heatmaps of different models, respectively. Note: Abbreviations are defined in the legend of Fig. 2.

However, the myeloid blast category exhibited the lowest accuracy at only 63.33 %, with frequent misclassifications into lymphoid precursor cells and mature lymphocytes.

Fig. 5(e) and (f) illustrate that in data-level fused GradCAM++ heatmaps of the 1D-GoogLeNet model, mature lymphocytes, basophils, and eosinophils exhibited similar spectral attention regions, concentrated primarily within the 450–550 nm band, closely matching those identified in single-spectral-data models. This consistency suggests that this wavelength region plays a crucial role in classification decisions. In contrast, attention regions for other cell types differed from single-spectral-data models but retained partial overlaps. Furthermore, monocytes cells showed markedly reduced or nearly absent attention regions in the phenotypic feature dimension, explaining the decreased accuracy of monocytes classification in fusion models. For other cells, phenotypic feature attention regions became more dispersed compared to single-modality models, possibly reflecting attention redistribution towards more generalized and supportive features due to competitive

multi-source information integration. For the 1D-ResNet model, GradCAM++ heatmaps in data-level fusion showed scattered attention regions in spectral data, with substantial overlap among different cell types. In phenotypic data, attention regions for lymphoid precursor cells, mature lymphocytes, and myeloid blasts became nearly mutually exclusive compared to single-modality models, indicating that fusion models dynamically adjusted attention weights, prioritizing highly discriminative features and suppressing features that were prominently utilized in single-phenotypic-feature models.

3.4. Classification models based on feature-level fusion

As illustrated in Fig. 6(a–d), the AUC values for most cell types across the four feature-level fusion models were generally higher compared to those obtained using data-level fusion. This observation suggests that feature-level fusion can more effectively capture features closely related to cell-type discrimination, thereby demonstrating greater robustness

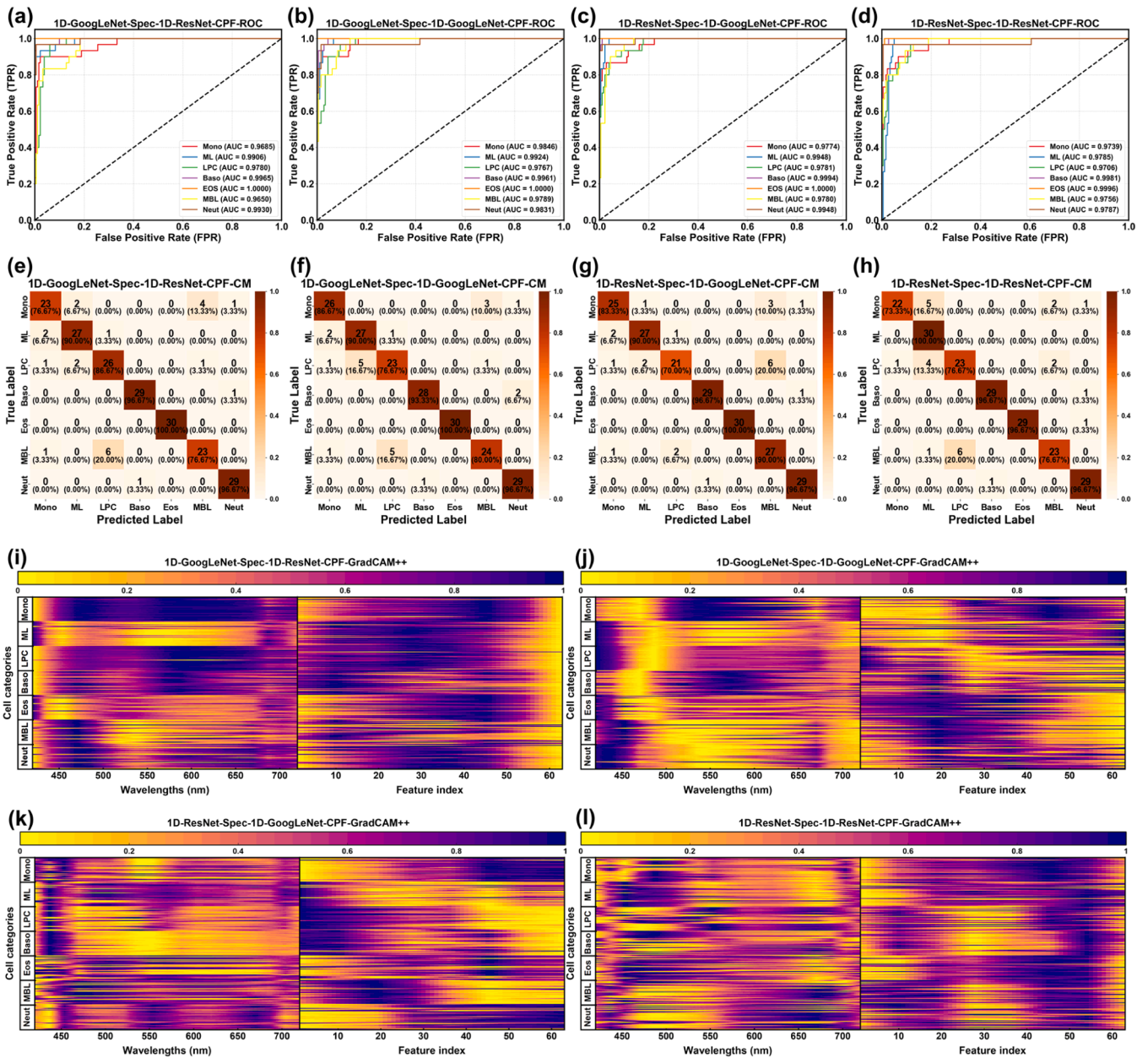


Fig. 6. Classification performance and visualization of models based on feature-level fusion strategy, (a–d), (e–f), and (i–l) present the multiclass ROC curves, confusion matrices, and GradCAM++ heatmaps of different models, respectively. Note: Abbreviations are defined in the legend of Fig. 2.

and practical utility in classification tasks. Notably, eosinophils consistently exhibited the highest AUC values (approaching or equaling 1.0000) among the four models. This superior performance likely arises from the distinct spectral signatures of eosinophils compared to other cell types, enabling accurate feature extraction and classification by all models. Furthermore, these models also exhibited excellent performance for other cell types, consistently achieving AUC values exceeding 0.9700, indicating strong discriminatory capabilities.

Fig. 6(e–h) further demonstrate that, compared to data-level fusion, feature-level fusion models yielded superior performance across most cell types, underscoring their advantages in exploiting complementary features and adapting effectively to integrated representations. Specifically, Fig. 6(f) indicates that the classification accuracy for monocytes reached 86.67 %, significantly higher than accuracies obtained with data-level fusion (63.33 % for 1D-GoogLeNet and 73.33 % for 1D-ResNet). Additionally, accuracies for basophils and mature lymphocytes were high, reaching 93.33 % and 90 %, respectively. Moreover,

the model presented in Fig. 6(h) achieved a classification accuracy of 100 % for mature lymphocytes, surpassing the best result (93.33 %) observed in data-level fusion. Additionally, feature-level fusion models consistently maintained 100 % accuracy in eosinophil identification, aligning with the data-level fusion results and highlighting the clear discriminative characteristics of eosinophils. Although some variability remained for certain cell types, such as lymphoid precursor cells, the overall findings indicate that feature-level fusion significantly enhances model discriminative power compared to data-level fusion strategies.

As depicted in Fig. 6(i–l), GradCAM++ visualizations under feature-level fusion indicate that the 1D-GoogLeNet architecture showed generally high consistency in activation regions, albeit with certain local differences. Specifically, within spectral data, the highlighted regions for most cell types (except lymphoid precursor cells) were spatially consistent, differing primarily in activation extent. In the phenotypic feature data, specific feature intervals (30–60 for monocytes, 1–20 for lymphoid precursor cells, and 5–30 for myeloid blasts) were identified

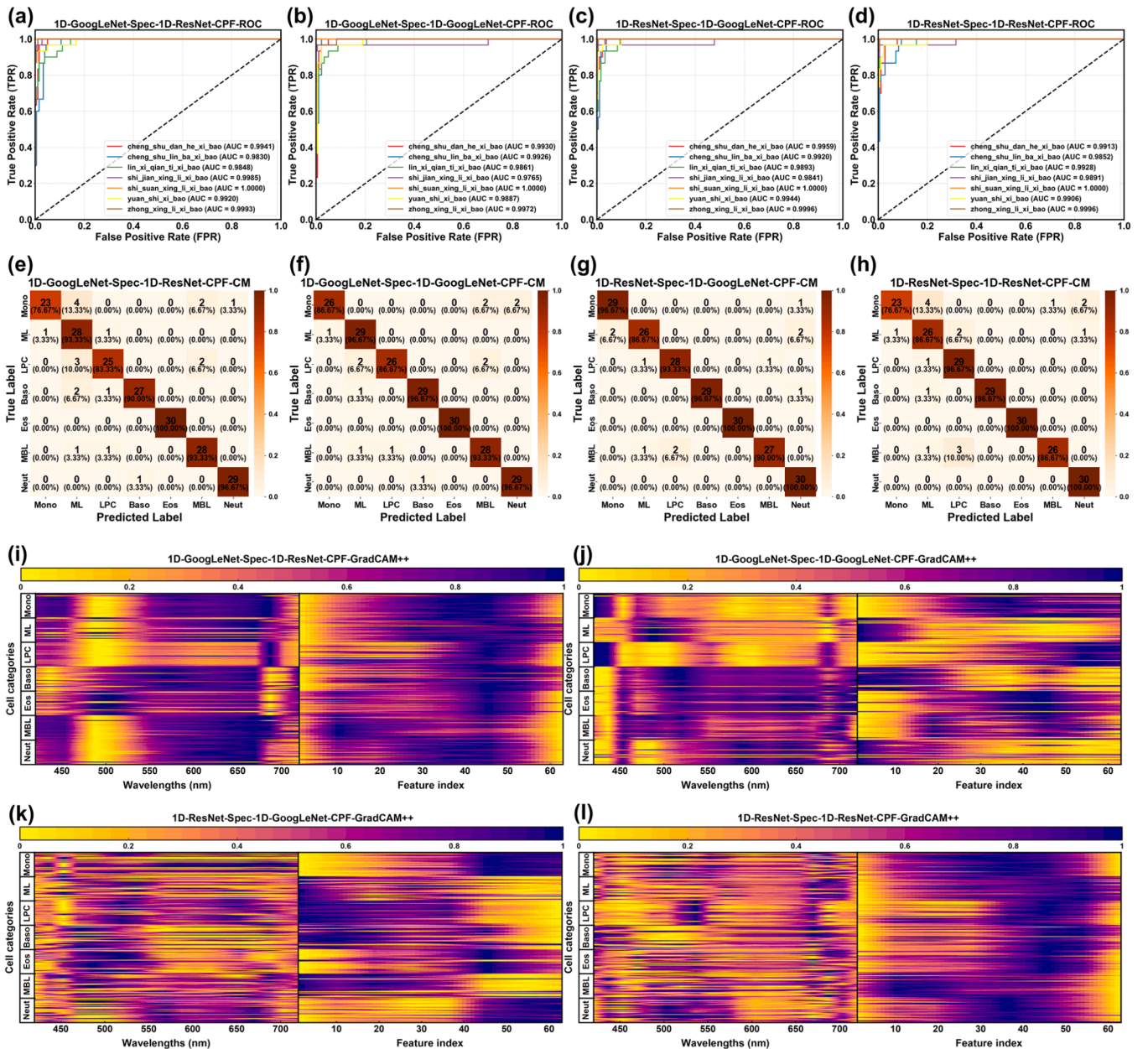


Fig. 7. Classification performance and visualization of models based on decision-level fusion strategy, (a–d), (e–f), and (i–l) present the multiclass ROC curves, confusion matrices, and GradCAM++ heatmaps of different models, respectively. Note: Abbreviations are defined in the legend of Fig. 2.

as significant discriminative regions. By contrast, the activation regions derived from the 1D-ResNet architecture exhibited more dispersed distributions across both spectral and phenotypic data, covering broader areas. This resulted in some convergence of highlighted regions among certain cell types. These observations suggest that the 1D-GoogLeNet architecture possesses superior capabilities in selecting and compressing discriminative features, while the 1D-ResNet architecture emphasizes global feature generalization. Additionally, the findings reveal that certain cell types exhibit structurally robust, modality-consistent discriminative features.

3.5. Classification models based on decision-level fusion

Fig. 7(a–d) show that among the three fusion strategies evaluated, decision-level fusion produced ROC curves reflecting the highest classification performance, with AUC values exceeding 0.9800 for all cell types and approaching or reaching 1.0000 for specific categories (e.g., eosinophils and neutrophils). This demonstrates exceptional sensitivity and specificity, highlighting the superior classification capability enabled by decision-level fusion. In comparison, data-level fusion also yielded relatively high AUC values but exhibited slightly lower performance for certain cell types such as myeloid blasts and lymphoid precursor cells, alongside ROC fluctuations at lower false-positive rates (FPRs). These observations indicate that direct concatenation of multi-modal data at the data level may introduce redundancy and noise, impairing model generalizability. Feature-level fusion, involving feature concatenation, improved upon data-level fusion with most categories achieving AUC values above 0.9700, though minor fluctuations persisted in some ROC curves.

Fig. 7(e–h) reveal that, within the optimal decision-level fusion models, classification accuracies exceeded 90.00 % for all WBC subtypes except mature lymphocytes. Notably, accuracies for monocytes, basophils, eosinophils, and neutrophils reached 96.67 %, 96.67 %, 100.00 %, and 100.00 %, respectively. Importantly, monocytes, lymphoid precursor cells, and myeloid blasts, which previously demonstrated suboptimal performance under data-level fusion (highest accuracies of 73.33 %, 80.00 %, and 73.33 %) and moderate improvements under feature-level fusion (86.67 %, 86.67 %, and 90.00 %), achieved further accuracy enhancements through decision-level fusion (96.67 %, 93.33 %, and 90.00 %). This marked improvement underscores that decision-level fusion, through comprehensive weighting of multimodal feature representations and heterogeneous model decision boundaries, effectively exploits complementary information from spectral and phenotypic modalities, significantly enhancing discriminative power, robustness, and generalization performance.

GradCAM++ visualization results (Fig. 7(i–l)) under decision-level fusion indicate consistent feature-highlighted regions across different modalities for the 1D-GoogLeNet model. Within spectral data, significant regions overlapped substantially across most cell types; for instance, monocytes and lymphoid precursor cells prominently focused on the 420–450 nm band, basophils and eosinophils on the 450–670 nm band, and neutrophils separately on 440–455 nm and 550–670 nm bands. This confirms the model's effectiveness in extracting representative discriminative bands. Regarding phenotypic feature data, attention regions were consistent across most cell types, except lymphoid precursor cells and neutrophils, which displayed mutually exclusive highlighted areas. Conversely, the 1D-ResNet model exhibited more dispersed attention distributions in the spectral modality, covering broader wavelength ranges, indicating global feature extraction tendencies. For phenotypic data, decision-level fusion models based on 1D-ResNet exhibited nearly identical highlighted regions across different combinations, with high overlap among cell types, reflecting robust and generalized feature extraction. Additionally, decision-level fusion GradCAM++ visualizations closely resembled those from single-modality models, likely because decision-level fusion performs simple weighted aggregation of final logits, preserving each branch's original

feature extraction and gradient backpropagation patterns.

3.6. Performance comparison of multi-level fusion models

As shown in Fig. 8, a systematic comparison of models employing various fusion strategies (data-level, feature-level, and decision-level) on the dual-modal (spectral–phenotypic) dataset of seven WBC types demonstrated that decision-level fusion achieved the most significant performance enhancement. Specifically, the optimal model, which combines the 1D-ResNet architecture for spectral data and the 1D-GoogLeNet architecture for phenotypic features and integrates them via weighted probability averaging at the decision output, consistently outperformed other strategies, attaining an accuracy of 94.76 %, precision of 94.91 %, recall of 94.76 %, and F1-score of 94.75 %.

Analyzing each fusion level separately, the single-modal baseline revealed that the 1D-ResNet model applied to spectral data (1D-ResNet-Spec) yielded the highest accuracy of 89.52 %, underscoring the enhanced capacity of residual structures for deep extraction of high-dimensional continuous spectral features. Conversely, the 1D-GoogLeNet model on phenotypic features (1D-GoogLeNet-CPF) performed best, achieving an accuracy of 89.05 %, indicating that multi-scale convolution via Inception modules effectively captured discrete data characteristics. However, each single-modal branch was limited by incomplete information from its respective modality, resulting in overall recall and F1-scores constrained within the 85–90 % range.

Data-level fusion failed to provide performance gains. Its optimal model (1D-ResNet-Spec-CPF) exhibited a marked accuracy reduction to 85.24 %, corresponding to a decline of approximately 4.3–4.7 % across all performance metrics. This outcome suggests that direct concatenation at the raw data level exacerbates noise amplification and feature redundancy, hindering convolutional kernels from effectively capturing critical information. Feature-level fusion exhibited performance comparable to the single-modal baseline. The best-performing model (1D-ResNet-Spec-1D-GoogLeNet-CPF) matched the baseline accuracy of 89.52 %, with only slight decreases in precision by 0.34 % and in F1-score by 0.23 %. This finding indicates that embedding-space concatenation following individual network feature extraction reduces low-level noise interference but fails to yield substantial improvements due to challenges in aligning heterogeneous distributions within high-dimensional embeddings.

Decision-level fusion, however, achieved a decisive breakthrough. Compared to single-modal baselines, the optimal decision-level fusion model (1D-ResNet-Spec-1D-GoogLeNet-CPF) improved all metrics by approximately 4.7–5.3 %. This highlights that independently learning modality-specific feature spaces prior to high-level complementary decision integration maximizes both the global abstraction capabilities of residual networks for continuous spectral data and the multi-scale, fine-grained extraction advantages of Inception-based networks for discrete data details. Consequently, this strategy delivers robust and balanced classification outputs, significantly enhancing model robustness and generalization capability.

In summary, the effectiveness of multimodal fusion strongly depends on the fusion hierarchy: data-level fusion frequently introduces noise overshadowing feature complementarity; feature-level fusion offers minimal or marginal improvements; whereas decision-level fusion, by minimizing cross-modal interference, achieves maximal performance gains, thereby establishing itself as the optimal strategy for combined spectral–phenotypic classification tasks.

3.7. Differentiation of normal, AML, and ALL blood smears

Leveraging the optimal decision-level fusion model (1D-ResNet-Spec-1D-GoogLeNet-CPF) identified through the multi-level model performance comparison, we further explored its potential for differentiating normal, AML, and ALL blood smears by quantifying the proportion of various WBC subtypes.

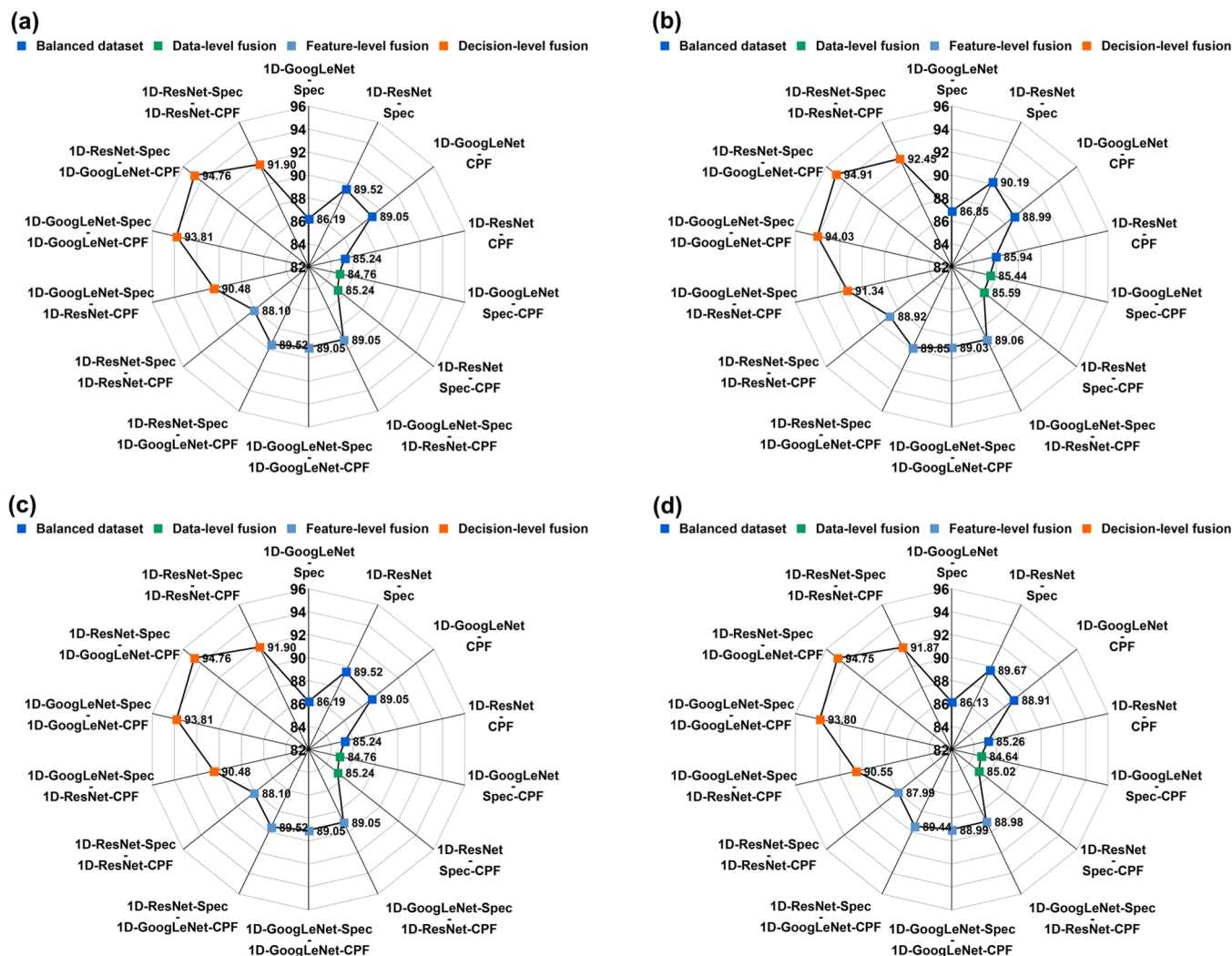


Fig. 8. Radar Charts Comparing Classification Performance of Different Models, (a–d) show accuracy, precision, recall, and F1-score, respectively.

An external validation cohort of 15 blood smear samples (five normal, five AML, and five ALL samples) was collected for model evaluation. Fig. 9 illustrates classification results and performance metrics of the decision-level fusion model on representative blood smears (see [supplementary material](#) for additional cases). Specifically, Fig. 9(a) represents an ALL smear with a significantly elevated proportion of lymphoid precursor cells (61.62 %), exceeding the 20 % threshold, consistent with classical ALL diagnostic criteria. Fig. 9(b) depicts an AML smear with a markedly increased myeloid blast proportion (44.38 %), surpassing 20 % and aligning with typical AML characteristics. Fig. 9(c) corresponds to a normal blood smear, exhibiting balanced distributions of cell types and notably low proportions (1.25 %) of both myeloid blasts and lymphoid precursor cells, reflecting normal peripheral blood WBC profiles.

Fig. 9(d) presents the confusion matrix for classification accuracy across all cells in the 15 externally validated blood smears. The model achieved an overall cellular-level classification accuracy of 92.90 %, demonstrating robust performance, particularly for mature lymphocytes, lymphoid precursor cells, basophils, eosinophils, myeloid blasts, and neutrophils, each exceeding 90 % accuracy. Furthermore, after quantifying cell proportions within each smear and applying diagnostic standards, the final diagnostic accuracy for differentiating normal, AML, and ALL smears reached 100 %.

In conclusion, the developed model successfully differentiated normal, AML, and ALL blood smears by quantifying proportions of

various cell types, providing a valuable reference for hematologic disease diagnosis.

4. Discussion

In this study, we developed an automated diagnostic framework integrating MHSI with deep learning for accurate differentiation among normal, AML, and ALL blood smears by quantitatively analyzing the proportions of different WBC types. This approach leveraged both spectral information and phenotypic features from individual cells, incorporated a WGAN-GP to address class imbalance, and systematically explored multiple data fusion strategies. In external validation, the model achieved a cellular-level classification accuracy of 92.90 % and a smear-level (based on cellular proportions) diagnostic accuracy of 100 %. These results underscore the substantial potential of multimodal data fusion and biologically-informed diagnostic criteria for advancing computational hematopathology.

To overcome the limitations of traditional leukemia classification methods regarding model interpretability and quantitative analysis of cell proportions, we constructed an automated framework combining MHSI and deep learning. Unlike conventional “black-box” approaches relying on entire images or single-modality data, our method takes individual cells as analytical units and integrates WHO diagnostic criteria, enabling precise quantification and automated identification of blast cell proportions for the first time. By quantitatively assessing the

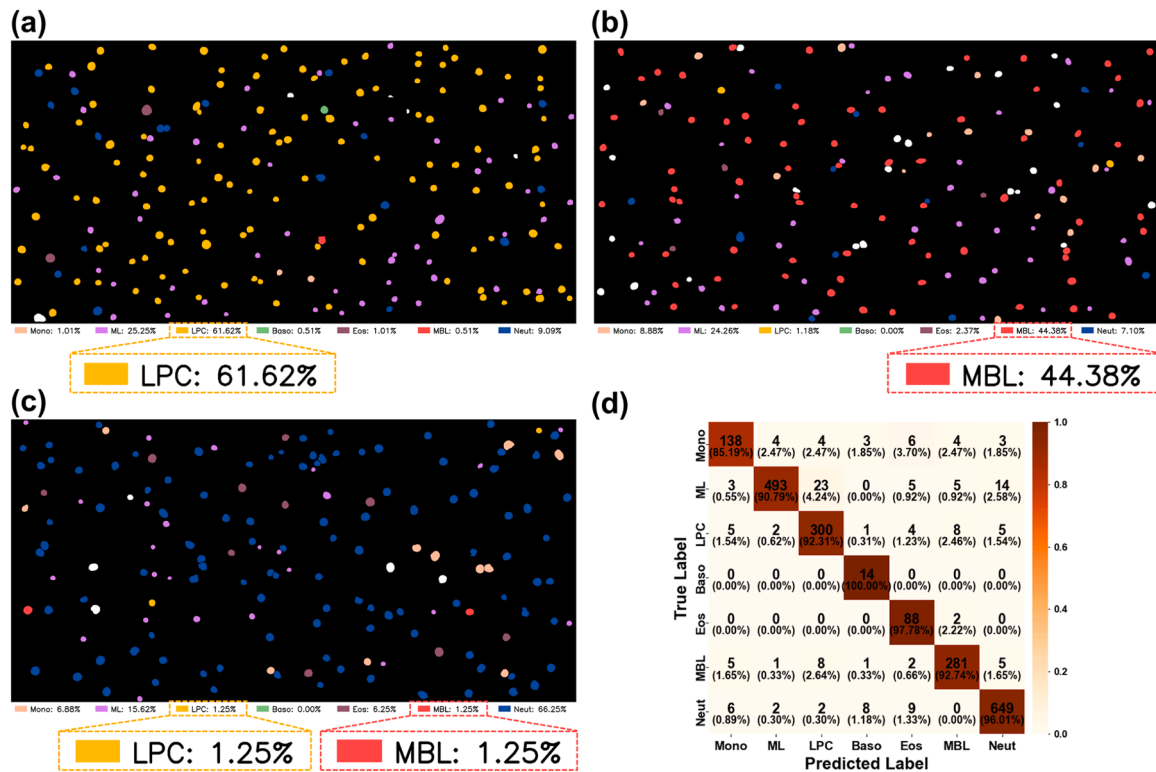


Fig. 9. Recognition Results and Classification Performance of the Optimal Model on Different Types of Blood Smears, (a–c) show the recognition results of ALL, AML, and normal blood smears, respectively. (d) Confusion matrix of the model's classification accuracy for all cells in 15 blood smears. Note: Abbreviations are defined in the legend of Fig. 2.

proportions of myeloid blasts and lymphoid precursor cells within blood smears, the model directly emulates diagnostic logic employed by hematopathologists, significantly enhancing clinical utility. The introduction of MHSI provided complementary sources of information, where spectral features reflect biochemical composition and extracted phenotypic features reveal morphological and structural differences among cells. This dual-modal information not only improved model discrimination but also enhanced the biological interpretability of Grad-CAM++ visualization results.

In terms of multimodal information fusion strategies, we systematically compared data-level, feature-level, and decision-level fusion methods. The decision-level fusion strategy demonstrated the best performance, significantly outperforming the other fusion approaches. This indicates that independent modeling in respective feature spaces followed by weighted probability integration at the output level effectively suppresses noise interference and fully exploits each network architecture's strengths. Specifically, the 1D-ResNet excels at extracting deep global features from continuous spectral data, whereas the 1D-GoogleNet demonstrates superior capability in handling discrete, multi-scale phenotypic features. Furthermore, to alleviate class imbalance, we employed a WGAN-GP for data augmentation.

It is worth emphasizing that this study provides a feasible technical route toward standardizing and automating the hematopathology diagnostic workflow. In current clinical practice, peripheral blood smear evaluation relies on laboratory personnel and hematopathologists to examine individual WBCs under the microscope, identify blasts/immature precursor cells, and estimate their proportion. This process is labor-intensive, time-consuming, and difficult to fully standardize. Multi-center investigations have reported that approximately 16.2 % of routine complete blood count (CBC) samples in clinical laboratories (and up to 26.7 % in some institutions) still require manual peripheral smear review [30]. Once a case is escalated for specialist hematopathologist review with an interpretive report, the turnaround time

can extend to 2–4 days [31]. This step also incurs additional professional service costs and is constrained by the limited availability of subspecialty expertise. At the same time, even among experienced hematopathologists, inter-observer variability persists in the identification and enumeration of specific WBC subtypes, particularly blasts and precursor cells, underscoring the intrinsic subjectivity of morphology-based diagnosis [32]. By contrast, the framework proposed in this study can automatically perform single-cell segmentation, cell-type classification, and quantification of blast/precursor cell proportions. In external validation, it achieved 92.90 % accuracy at the cellular level and 100 % accuracy at the smear level, providing consistent diagnostic assignments across blood smears. These findings suggest that the proposed method has the potential to reduce the time burden, personnel cost, and observer-dependent variability associated with manual smear review, while maintaining diagnostic consistency.

Despite achieving promising classification performance, this study still has several limitations that require further attention. First, external validation was conducted on only 15 blood smears, a relatively small and homogeneous sample set. Larger-scale, multi-center datasets are necessary to further validate the generalization of the algorithm under different staining protocols and imaging platforms. Second, the present MHSI system remains relatively slow in terms of data acquisition. To address this bottleneck, future work will focus on two directions. At the hardware level, parallel acquisition strategies such as snapshot hyperspectral imaging or multi-field parallel capture could be introduced, enabling the system to obtain spatial and spectral information simultaneously in a single exposure rather than relying on sequential band-by-band and region-by-region scanning, thereby substantially shortening the hyperspectral acquisition time per cell. In addition, the current workflow of manually selecting individual regions of interest could be replaced by an automated procedure in which the entire smear is first rapidly captured using conventional brightfield/color imaging, white blood cell-dense regions are automatically identified and their

coordinates recorded, and only these targeted regions are subsequently acquired at hyperspectral resolution. Lastly, prospective validation within real clinical workflows, along with cost-effectiveness analyses involving manual smear examination, is required to assess the feasibility and clinical value of the proposed system in practical diagnostic scenarios.

5. Conclusion

This study introduces an automated diagnostic framework combining MHSI with deep learning, achieving high-precision classification of normal, AML, and ALL blood smears by accurately quantifying the proportions of different cell types. Using MHSI, we simultaneously acquired spectral and color image from blood smears and performed high-quality single-cell segmentation via the U²-Net model. Subsequently, we extracted spectral data spanning 420–720 nm and 63 phenotypic features for downstream modeling. To address data imbalance, we employed a WGAN-GP generative adversarial network for data augmentation. Regarding model construction, two one-dimensional convolutional neural networks, 1D-GoogLeNet and 1D-ResNet, were developed and systematically evaluated using three multimodal fusion strategies: data-level, feature-level, and decision-level fusion. The decision-level fusion model, specifically the 1D-ResNet-Spec-1D-GoogLeNet-CPF configuration, exhibited the best performance, achieving an internal test set accuracy of 94.76 %. In external validation involving 15 independent clinical samples, the cellular-level classification accuracy was 92.90 %, while smear-level diagnostic accuracy (based on WHO-defined cell proportions) reached 100 %. This study integrated MHSI with multimodal deep learning for leukemia type classification, demonstrating its strong potential in both single-cell recognition and smear-level diagnosis. The proposed approach holds promise for providing reliable technological support for intelligent triage and standardized auxiliary diagnosis of leukemia.

CRedit authorship contribution statement

Hai Yin: Conceptualization, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Li Li:** Data curation, Formal analysis, Methodology. **Gehua Huang:** Investigation, Supervision, Validation. **Qihang Yang:** Investigation, Software, Supervision, Validation, Visualization. **Danhui Gan:** Data curation, Formal analysis, Project administration, Resources, Validation. **Siqi Zhu:** Resources, Software, Supervision, Validation. **Furong Huang:** Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.snb.2025.139048](https://doi.org/10.1016/j.snb.2025.139048).

Data availability

Data will be made available on request.

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